PROMOSON POUDEL

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PROFESSIONAL SUMMARY :

Accomplished and versatile Senior Data Engineer with 6+ years of experience architecting scalable, secure, and high performance data platforms across healthcare, finance, insurance, and retail domains. Strong expertise in designing modern cloud native **Lakehouse Architecture**, enterprise **Data Warehouses**, and analytics platforms using **Azure Data Factory**, **Azure Databricks**, **Snowflake**, **Delta Lake**, **AWS Glue**, **Amazon S3**, **Amazon Redshift**, **GCP BigQuery**, **Dataflow**, **Pub/Sub**, and **Dataproc**. Skilled in building metadata driven **ETL/ELT Frameworks**, batch and real time streaming pipelines, **Data Integration** solutions, and optimized **Data Models** that improve scalability, governance, **Data Quality**, **Data Lineage**, and regulatory compliance aligned with **HIPAA**, **GDPR**, and enterprise security standards.

Experienced in developing AI ready data platforms and supporting **Generative AI** and **RAG Pipeline** use cases by preparing curated, governed, and analytics ready datasets for **Machine Learning**, **Vector Search**, **Semantic Retrieval**, advanced analytics, and executive reporting. Proficient in **Python**, **SQL**, **PySpark**, **Apache Spark**, **Kafka**, **Airflow**, **dbt**, **Terraform**, **Git**, **Azure DevOps**, **CI/CD Automation**, **API Integrations**, and cloud native monitoring tools to deliver reliable production grade data solutions. Adept at collaborating with data science, **BI**, **DevOps**, security, and business teams to deliver trusted datasets, **Power BI** and **Tableau** dashboards, machine learning workflows, and mission critical decision making solutions at enterprise scale.

TECHNICAL SKILLS:

- **Cloud Platforms & Services:** Azure (Azure Data Factory, Azure Databricks, Synapse Analytics, ADLS Gen2, Event Hubs, Event Grid, Azure Functions, Azure SQL Database, Blob Storage, Azure Monitor, Log Analytics, Key Vault, Managed Identity, Microsoft Purview), AWS (Glue, S3, Redshift, Lambda, Kinesis, EC2, IAM, CloudWatch, Lake Formation, Athena, Step Functions, EventBridge, RDS, DynamoDB), GCP (BigQuery, Cloud Storage, Pub/Sub, Dataproc, Dataflow, Cloud Composer, Cloud Functions, Cloud IAM, Cloud Logging, Cloud Monitoring)
- **Data Warehousing:** Snowflake, Amazon Redshift, Azure Synapse Analytics, Google BigQuery, SQL Server, Oracle, Teradata
- **Snowflake Features:** Streams, Tasks, External Tables, Secure Data Sharing, Dynamic Data Masking, Zero Copy Cloning, Time Travel, Snowpipe, Materialized Views, Query Optimization
- **Lakehouse Platforms:** Azure Databricks, Delta Lake, Delta Live Tables, Databricks Workflows, Unity Catalog, Databricks Repos, Feature Store, Medallion Architecture, Bronze Silver Gold Layers
- **ETL and ELT Tools:** Azure Data Factory, AWS Glue, Apache Airflow, MWAA, dbt, Informatica PowerCenter, SSIS, SQL Based ELT, CDC, Autoloader, Glue Workflows
- **Data Orchestration:** Apache Airflow, Azure Data Factory Pipelines, Databricks Workflows, AWS Step Functions, EventBridge, Azure Event Grid, Cloud Composer
- **Programming Languages:** Python, SQL, PySpark, Spark SQL, T SQL, PL/SQL, Bash, Shell Scripting, Scala
- **Big Data Processing:** Apache Spark, PySpark, Spark SQL, Databricks, Dataproc, AWS Glue Spark, EMR, Distributed Processing, Partitioning, Caching, Performance Tuning
- **Streaming and Real Time Processing:** Apache Kafka, Amazon Kinesis Data Streams, Azure Event Hubs, Google Pub/Sub, Spark Structured Streaming, Micro Batch Processing, Event Driven Architecture
- **Databases:** PostgreSQL, MySQL, SQL Server, Oracle, Teradata, MongoDB, Redis, DynamoDB, Amazon RDS, Azure SQL Database
- **Data Modeling:** Dimensional Modeling, Star Schema, Snowflake Schema, SCD Type 2, Slowly Changing Dimensions, Data Vault Basics, Fact and Dimension Tables, Schema Evolution
- **Data Architecture:** Lakehouse Architecture, Delta Architecture, Medallion Architecture, Data Lake, Enterprise Data Warehouse, Metadata Driven Design, Reusable Pipeline Frameworks
- **Data Governance and Security:** Microsoft Purview, Unity Catalog, AWS Lake Formation, Data Cataloging, Data Lineage, RBAC, IAM Policies, PII Protection, Encryption, Dynamic Masking, Secure Data Sharing
- **Data Quality and Validation:** Great Expectations, Data Profiling, Data Reconciliation, Schema Validation, SLA Monitoring, Exception Handling, Audit Logging, Data Quality Rules

- **DevOps and CI/CD:** Azure DevOps Pipelines, GitHub Actions, Git, Databricks Repos, CI/CD Automation, Release Management, Version Control, Deployment Automation
- **Infrastructure as Code:** Terraform, IAM/RBAC Policies, Cloud Resource Provisioning, Environment Configuration, Reusable Infrastructure Modules
- **Monitoring and Observability:** Azure Monitor, Log Analytics, AWS CloudWatch, Cloud Logging, Cloud Monitoring, Databricks Jobs Monitoring, SLA Alerts, Custom Dashboards
- **AI, ML and GenAI Data Support:** Feature Engineering Pipelines, Feature Store, RAG Pipelines, Vector Search, Semantic Retrieval, Embeddings Data Preparation, ML Ready Datasets, Model Input Pipelines
- **BI and Reporting Tools:** Power BI, Tableau, Looker Studio, Excel, Executive Dashboards, KPI Reporting, Self Service Analytics
- **APIs and Data Integration:** REST APIs, JSON Parsing, XML Parsing, API Based Ingestion, File Based Ingestion, Batch Integration, Real Time Integration
- **File Formats:** Parquet, Avro, ORC, CSV, JSON, XML, Delta Format
- **Operating Systems and Tools:** Linux, Unix, Windows, Jira, Confluence, Agile Scrum, ServiceNow

PROFESSIONAL SUMMARY:

Senior Data Engineer
Elevance Health – Indianapolis, IN **May 2024 – Present**

- Served on Elevance Health's Enterprise Healthcare Data Modernization team, designing a scalable healthcare lakehouse platform using **Azure Data Lake Gen2**, **Azure Databricks**, **Delta Lake**, and **Snowflake** to unify claims, eligibility, provider, and clinical quality datasets supporting **120M+ claims records daily**.
- Re engineered fragmented ingestion workflows into a standardized **Bronze**, **Silver**, and **Gold** architecture using **Databricks Autoloader**, **PySpark**, **CDC**, **Delta Lake**, and **dbt**, improving data freshness and analytics readiness by **35%**.
- Built reusable metadata driven **ETL/ELT** pipelines in **Azure Data Factory** and **Azure Databricks** to support batch, incremental, and partner data feeds with improved scalability and operational consistency.
- Designed secure data architecture for **PHI**, **PII**, and healthcare sensitive workloads by implementing **RBAC**, **Managed Identities**, **Azure Key Vault**, encryption, masking policies, and access controls aligned with **HIPAA**standards.
- Optimized large scale **PySpark** workloads in **Azure Databricks** by improving partitioning, shuffle management, caching, and Delta table design, reducing pipeline runtime by **35 to 40%** and improving compute efficiency.
- Enhanced **Snowflake** performance and cost governance through warehouse right sizing, clustering, query profile analysis, materialized views, secure data sharing, row level security, and dynamic masking, reducing annual warehouse spend by **25 to 30%**.
- Partnered with clinical quality, risk adjustment, actuarial, finance, and BI teams to deliver curated datasets for **HEDIS**, **RAF scoring**, regulatory reporting, executive dashboards, and operational analytics.
- Developed semantic marts and dimensional data models using **dbt**, **SQL**, **Snowflake**, and **Delta Lake** to support self service reporting, KPI tracking, claims trend analysis, and provider performance reporting.
- Supported **AI**, **machine learning**, and **RAG pipeline** initiatives by preparing governed, curated, and feature ready datasets for model training, semantic search, vector based retrieval, anomaly detection, and predictive analytics.
- Collaborated with data science teams to build feature engineering pipelines and ML ready datasets using **Databricks Feature Store**, **PySpark**, **Delta Lake**, and **Snowflake** for risk scoring and member segmentation use cases.
- Designed cross cloud ingestion pipelines to securely integrate partner and provider datasets from **AWS S3** into Azure using automated validation, schema checks, and Databricks based transformation workflows.
- Implemented real time validation workflows using **Azure Event Grid**, **Azure Functions**, and **Databricks** to detect anomalies in eligibility, claims, and provider data before downstream reporting impact.
- Improved production reliability by redesigning orchestration logic in **Azure Data Factory**, adding dependency aware triggers, retry mechanisms, data quality gates, and SLA based monitoring, reducing recurring pipeline failures by **40%**.
- Automated infrastructure provisioning and deployments using **Terraform**, **Azure DevOps Pipelines**, **Git**, and **CI/CD**, ensuring consistent environment configuration across development, testing, and production while reducing manual configuration drift.
- Mentored data engineers and analysts through architecture reviews, code reviews, **PySpark** optimization sessions, data modeling guidance, and governance best practices to improve delivery quality and platform maturity.

Key Technologies: Azure Data Factory, Azure Databricks, Azure Synapse Analytics, Snowflake, Delta Lake, Azure Data Lake Storage Gen2, Azure Monitor, Azure DevOps, Azure Functions, Azure Key Vault, Terraform, Power BI, ADF, Great Expectations, RBAC, Managed Identity, AWS S3, GCP, PySpark, dbt

Data Engineer
MetLife – New York City, NY **Mar 2023 – Apr 2024**

- Supported MetLife's insurance data platform team in modernizing policy, claims, underwriting, actuarial, and customer analytics data pipelines across cloud based data lake and warehouse environments.
- Developed ingestion pipelines for policy, claims, billing, and underwriting datasets using **AWS Glue**, **Amazon S3**, **PySpark**, and **SQL**, improving data availability for reporting and downstream analytics teams.
- Re engineered legacy **SSIS** and SQL based workflows into scalable **PySpark** transformation jobs with incremental load logic, partitioning, and reusable processing patterns for insurance data workloads.
- Built and maintained structured data lake zones for raw, enriched, and curated datasets, applying file organization, compression, partitioning, and metadata standards to support better data discovery and performance.
- Created transformation logic in **Databricks** using **Delta Lake**, **PySpark**, schema enforcement, and merge patterns to support actuarial analysis, claims reporting, and regulatory data needs.
- Assisted with **Snowflake** warehouse optimization by tuning SQL queries, reviewing query profiles, supporting warehouse right sizing, and improving performance for reporting and analytics workloads.
- Developed modular **dbt** models to prepare curated data marts for claims analytics, policy lifecycle reporting, underwriting performance, and actuarial business reporting.
- Supported event driven data processing for priority claims and fraud related data using **Kinesis**, **Lambda**, and cloud based ingestion patterns, improving data readiness for investigation workflows.
- Created SQL based datasets and reporting layers for **Power BI**, **Tableau**, and business dashboards, helping claims, underwriting, actuarial, and leadership teams track KPIs and operational trends.
- Assisted in building governed access patterns using **IAM**, role based access controls, masking rules, encryption standards, and audit logging to protect sensitive policyholder and claims data.
- Supported data quality checks using **SQL**, **Python**, and validation rules to identify missing records, schema mismatches, duplicate data, and reconciliation issues across insurance data pipelines.
- Prepared clean and structured datasets to support **AI**, **machine learning**, fraud analytics, anomaly detection, and risk scoring use cases for claims and customer data analysis.
- Used **Terraform**, **Git**, and **CI/CD** pipelines to support repeatable deployments, version control, and controlled releases across development and production environments.
- Monitored data pipeline health using **CloudWatch**, logs, alerts, and operational dashboards, helping troubleshoot failed jobs, delayed loads, and production data issues.
- Collaborated with senior engineers, BI developers, analysts, and business stakeholders to review requirements, support data modeling, document pipeline logic, and deliver reliable analytics ready datasets.

Key Technologies: AWS Glue, Amazon S3, Kinesis, Lambda, Redshift, Athena, QuickSight, Snowflake, Databricks, Delta Lake, Apache Airflow, Terraform, CodePipeline, Azure Blob Storage, Google Cloud Storage, Lake Formation, dbt, PySpark, CI/CD

ETL Developer
Home Depot – Atlanta, GA **Sept 2019 – Feb 2023**

- Worked within Home Depot's Retail Data and Supply Chain Analytics team, supporting ETL development for sales, inventory, merchandising, pricing, and supplier datasets across store operations and e commerce reporting platforms.
- Developed and maintained ETL workflows using **Informatica PowerCenter**, **SSIS**, **SQL Server**, **Oracle**, **Teradata**, and flat file sources to support enterprise data warehouse and retail analytics requirements.
- Created source to target mappings, transformation logic, lookup rules, filters, joins, aggregations, and data validation checks to load structured retail data into centralized warehouse tables.
- Built SQL based transformation logic using **T SQL**, **PL/SQL**, stored procedures, views, and set based processing to support nightly batch loads for high volume POS, inventory, and promotional datasets.
- Supported performance tuning of ETL jobs by reviewing query execution plans, optimizing joins, improving indexing strategies, and assisting with partitioned table loads in **Teradata** and **SQL Server**.
- Enhanced existing **SSIS** packages by adding reusable variables, parameters, checkpoints, error handling, and dependency checks to improve batch processing stability.
- Developed and maintained **Informatica PowerCenter** mappings, sessions, and workflows for extracting data from relational databases, flat files, and operational retail systems.
- Implemented **SCD Type 1** and **SCD Type 2** logic for product, pricing, supplier, and store dimension tables to support historical tracking and accurate retail reporting.

- Created reconciliation scripts using **SQL** and **Shell Scripting** to compare source and target record counts, identify missing records, and validate data accuracy after ETL loads.
- Assisted in building star schema and snowflake schema structures for sales, inventory, merchandising, and supply chain reporting within enterprise data warehouse environments.
- Scheduled and monitored ETL jobs using **SQL Server Agent**, **Informatica Workflow Monitor**, and batch scripts, supporting timely completion of daily and weekly data refresh cycles.
- Troubleshot production ETL failures by reviewing logs, identifying data quality issues, coordinating with senior engineers, and applying fixes for recurring load and transformation errors.
- Supported report ready datasets for **SSRS**, **Crystal Reports**, **Power BI**, and business dashboards used by merchandising, inventory planning, finance, and operations teams.
- Used **TFS**, **Git**, and change management processes to manage ETL code versions, package migrations, deployment documentation, and release activities across environments.
- Collaborated with data modelers, QA analysts, business analysts, and senior data engineers to gather requirements, document mapping logic, validate test results, and deliver reliable analytics ready datasets.

Key Technologies: Informatica PowerCenter, SSIS, SQL Server, Oracle, Teradata, DB2, dbt, PL/SQL, Bash, Shell Scripting, Crystal Reports, SSRS, TFS, SCD, ETL, Data Warehousing, Star Schema, Performance Tuning

EDUCATION:

Bachelor of Science in Information System, University of Texas at Arlington, Arlington, TX

CERTIFICATIONS:

- Databricks Certified Data Engineer Associate
- Microsoft Certified: Azure Data Engineer Associate (DP 203)
- Snowflake SnowPro Core Certification